Lesson Aim

Explain the basic principles of wine making.

Introduction

Centuries ago the psalmist wrote "Wine maketh glad the heart of man". Wine, the fermented juice of freshly gathered grapes, has been made since before recorded history. Strictly speaking, wine can be made from nothing else but grapes. Hence “wines” made from other types of fruit are not true wines.

Essentially, winemaking is a combination of chemistry and microbiology. However, you do not need to be either a chemist or a microbiologist to be successful at making good wines, but it is important to adhere to the fundamentals and follow fixed procedures. This does not mean though that inconsistency in the final product is necessarily a bad thing. Each wine has an individual character and exhibits the personality and taste of its creator. Good wine is, however, not produced by chance.

Four major factors influence wine production:

- Soil from which the vine grows
- Climate specific to the site where grapes are grown
- Variety of grape used
- Inputs from the winemaker or **vigneron**.

Selection of the grapes used for winemaking greatly determines the characteristics of the final product. The viticulturist’s choices and management skills determine the quality of grape delivered to the winemaker. The vigneron selects the specific variety of grape for the wine he or she wishes to make. Basic flavours and characteristics are directly related to variety; the first choice bein of course, white or red selections. The features deriving from these fundamental inputs are unalterable.

Basic Production

The basic processes of winemaking can be summarised as follows:

1. Removal of unessential material such as stems and leaves
2. Grape crushing
3. Fermentation
4. Storage or bottling

The basic difference between red and white wines is that in the production of white wines, skins are removed prior to fermentation, whereas in the production of red wines, skins are fermented along with the juice.

Wine making is an accomplished art and the wine quality depends greatly on the skill of the winemaker. Good wine makers are always in demand.

An Overview of Winemaking Processes

1. Grapes are picked and brought to the press house as soon as possible after picking. They are fed into a crushing machine to extract the juice. The crusher also removes stems. Stems will increase the pH of the juice which, in turn, decreases colour intensity and fruitiness. Crushing is the process of mechanically breaking the grape skins in order to expose the juice to the yeast and permit alcoholic fermentation. Tannins will be extracted mainly from the grape skins, however, some stems may be left in or even added in small amounts to increase tannin levels (especially in reds).

2. The next stage in the process follows a different course according to the type of wine being made:

- White grapes have their skin removed.
- Red grapes being used to make red wine have their skins left in during fermentation (if skins were removed, red grapes would turn into white wine). Maceration is the process of letting crushed grapes soak in their juice for the purpose of extracting phenols from the solids. Reds wines obtain their distinctive ‘structure’ during this phase. They develop colour, flavour and texture. Full-bodied reds, such as Cabernet Sauvignons may be left to macerate for up to 21 days.
- With Rosé, skins from red grapes are removed a short time after fermentation starts.

3. Pressing is the process of juice extraction from the crushed grapes by pressure. Pressed white grapes produce a cloudy juice that will require clarification during the final stages of vinification. The juice derived from pressing is usually fermented separately.

4. Alcoholic fermentation. This stage is where the juice is converted into wine and hence is an important step in good winemaking. Yeasts will convert sugar into alcohol during fermentation so the strain of yeast used is vital. The winemaker needs to control factors such as sugar concentration, acidity and pH, temperature, SO_2 levels. Fermentation should be monitored regularly by assessing the changing sugar concentration (Brix) in the container. Secondary, or malolactic, fermentation is sometimes used after alcoholic fermentation.

5. Clarification removes cloudiness and haze from wine. A good wine should be free of suspended particles. Sedimentation and racking are traditional clarification methods. Fining agents and filtration are more often used in commercial winemaking for expedience.

6. Sometimes wine from two or more varieties of grape is blended to yield the final drinkable product. Blending has an extensive tradition and is responsible for the distinct flavour and characteristics of regional wines. This process is also called assemblage. Blending between vintages is used to ensure consistent, good quality product. Blending allow the winemaker to instil a good balance to the wine they produce. Well done properly, blending establishes a good balance of complexity of smell (bouquet) and taste, body, fruitiness, colour intensity, acidity, oak character and many other important characteristics.

7. Storage and barrelling can enhance the characteristics the winemaker has infused into the wine. Complexity of flavour and feel can be improved through oak barrelling. The main difference between storage in steel or glass containers versus barrels is the ve slow diffusion of air and evaporation that occurs through barrels. Some wines are actually fermented in barrels, resulting in complex flavours.

8. Bottling is the final stage in the winemaking process. Wine is bottled for the purpose of distribution for immediate consumption or for further ageing. The main goal of bottling is to prevent the wine from further exposure to air. The actual shape of the bottle can easily identify wines of particular types e.g. Burgundy-style, Champagne, Port. Wines to be aged are stored in green or brown-tinted bottles. Traditionally premium wines have been sealed with a natural cork, however, there are alternatives. Synthetic corks are available and screw caps that offer effective sealing are becoming more popular. Capsules are often placed over corked necks to retard mould formation and improve the look of the bottle.

9. Cellaring. Certain wines will be enhanced by time spent in storage. However, storage conditions will affect the wine’s ageing process. Cellaring conditions should provide a constant temperature between 10°C and 13°C, constant humidity between 55 and 75%, complete darkness, and a clean atmosphere, e.g. no strong smells.

Some Basic Production Principles

- Use only clean equipment and containers. Micro-organisms on the surfaces of equipment that has not been thoroughly washed and sterilised can severely affect the fermentation process. Conversely, leftover cleansing agents such as detergents and sterilents can affect the microbial balance of the wine.
- Begin with the best quality grapes that have reached a suitable degree of ripeness.
- Limit oxygen exposure to the fermenting juice.
- Keep the grapes, juice or wine below 15°C.
- Use quality yeast suitable for winemaking.

The activities of many different types of yeasts and bacteria needs to be controlled to ensure the final result is what you require.

Fermentation

Wild yeasts start the fermentation process, but they only produce about 4% alcohol and impart a yeasty, bitter taste to the wine. Wine yeasts then take over, changing the remaining sugar into alcohol (bringing the alcohol content to about 10%). If left unchecked, vinegar bacteria then take over, turning the wine into vinegar.

The winemaker must remove unwanted wild yeasts and vinegar bacteria if he or she is to control the fermentation process and produce quality wine. Potassium bisulphite is added to the must to kill unwanted microorganisms and prevent oxidation.

The sugars that provide the basic substrate for the reaction to occur are found in the grapes. Even though yeast is found naturally on the surface of grapes, “wine yeast” is added to the crushed grape pulp to produce the type of fermentation required for quality wine production. The important characteristic of yeasts is that they metabolise sugars to form alcohol without using oxygen. This process is also referred to as **anaerobic respiration**. Certain commonly found bacteria are also capable of anaerobic respiration, but with the undesirable result of acetic acid (vinegar) or other souring compounds. If present these bacteria will lead to unpleasar odours and tastes in the wine, hence, the importance of clean, uncontaminated equipment.

The speed of fermentation is dependent on temperature. In many commercial wineries, the temperature of the fermentation tanks is carefully controlled to produce steady temperatures for optimum fermentation. Yeasts only live in temperatures between 4°C an 32°C. The closer to 4°C, the slower they work; the closer to 32°C, the faster they work. White wines are generally fermented in temperatures in the range of 12° to 18°C; reds are fermented at higher temperatures.

A short fermentation period produces a light, inexpensive wine. A longer fermentation time produces a finer quality wine.

Process for Making White Wine

- Use red or white grapes.
- Crush grapes and remove stems and skins.
- Fermentation time is determined by the type of wine required:
    - Sweet wine is produced by stopping fermentation before all of the sugar is converted to acid.
    - Dry wine is produced by allowing all sugar to be converted to acid.
    - Sparking wine is made several ways; the most natural being to bottle the wine before fermentation is complete and allow fermentation to finish in the bottle.

Process for Making Red Wine

- Only use red or black grapes.
- Stems may or may not be removed when grapes are crushed.
- Grapes go into the fermenting tank with the skins (and sometimes stems).
- Wine which runs off freely is put into barrels and allowed to mature for varying periods.

- Skins and remaining moisture in the vat can be pressed to retrieve further dark and bitter wine. This can be blended with the wine which was run off earlier to give differing tastes.

Method for Home Wine Making

Good wine is certainly more than fermented grape juice. The complex flavours and aromas that are vital need to be extracted from the grapes.

Preparing to Ferment

1. Sugar is added. When this comes into contact with yeast it turns from a white powder into a fizzy gas (carbon dioxide), and a simple molecule called ethanol (alcohol). There is much debate as to the quantity of sugar added. It must be a minimum of 10% alcohol by volume which will produce the minimum percentage of alcohol to preserve the wine. On the other hand too much sugar will produce undrinkable syrup. The precise measurement of the amount of sugar is referred to as gravity and is measured using a hydrometer.

Wine is usually started with a gravity of between 1090 and 1100 (though some home winemakers start with a kilo of sugar for a gallon of must, and then add up to another 8 oz. later on). The first method gives more control over proceedings. More sugar is added to sweet wines than dry ones, and alcohol will usually end up at around 15%.

2. Tannins are added. This may be in tablet or liquid form.

3. It may be desirable to add some pectic enzyme. This gets rid of haze induced by pectin. It is available as a powder or a liquid.

4. Acid is added. If the finished wine is too acidic it will taste sharp. If it is too alkaline it will taste bland. Winemakers aim to be somewhere between the two extremes. The universal measure of acidity is parts per 100,000 of sulphuric acid (ppt).

Wine ranges from 3.0 ppt for a light dry wine to 4.5 ppt for a sweet full-bodied wine. Winemakers attempt to get the ppt within 0.1 ppt of where they want it by adding acid or calcium carbonate, to raise acidity or alkalinity respectively. There are a number of different acids available for this purpose, including:

- Citric
- Malic
- Tartaric
- Succinic
- Tannic

5. Once the must has cooled the yeast is added. Yeasts should be stored at temperatures that do not fluctuate too much, and they need to be fed with yeast nutrients or sugar. They also prefer a slightly acidic environment. There are many different types of yeas to choose from including standard wine yeasts and more specialist ones.

The whole mixture is now stirred and left at a temperature above 15 degrees C to allow it to ferment. The fermenting must is usually strained after a few days. Homemade wines are put in demijohns at this point and fitted with an airlock watch.

Fermentation

The mixture is left for weeks to months until the fermentation process stops. This can be determined by:

- A specific gravity of 1005 or less
- No bubbles in the airlock
- The wine begins to clear from the top

The wine is now racked (siphoned) leaving the gunk (lees) behind.

Clearing and Finishing

The aim is to now turn the siphoned wine into crystal clear wine that is bottled and labelled.

- Stabiliser and crushed Campden tablets may be added. After 24 hours no further fermentation will take place, and the wine will be stabilised.
- Wine finings can be added to remove any residue within the next week.
- The wine is then racked again.
- Taste the wine and adjust acidity/alkalinity levels.
- Filter the wine.
- Pour into wine bottles, cork and leave to mature.

Components of Grape Juice that are Important for Wine Production

Sugars – glucose, fructose
Organic acids – tartaric acid, malic acid, citric acid
Tannins – catechol, chlorogenic acid, caffeic acid
Amino acids and proteins
Minerals – phosphates, sulphates
Vitamins – B-group, ascorbic acid
Important volatile components that affect flavour and colour